

Writabrata Bhattacharya

Senior Applied Scientist | GenAI · LLMs · Speech AI · MLOps · Time-Series Forecasting

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Experience

Inception Institute of Artificial Intelligence, Senior Applied Scientist

Abu Dhabi, UAE
July 2024 – present
2 years 3 months

- **EnergyAI · ADNOC**

- Led design and development of advanced RAG systems for upstream oil & gas operations, handling structured and unstructured data to deliver context-aware responses
- Optimized RAG pipeline using Azure Document Intelligence for PDF parsing and dynamic semantic chunking, minimising context loss
- Fine-tuned Meta's LLaMA 70B using PEFT and full fine-tuning; performed LoRA fine-tuning on Qwen3-Embedding-8B with model merging (arXiv:2601.04720) for superior domain-specific retrieval
- Built end-to-end evaluation framework auditing all RAG stages (OCR, chunking, retrieval accuracy, generation fidelity) for systematic performance tracking
- Benchmarked state-of-the-art and open-source LLMs and embedding models on oil & gas domain data
- Developed a Chatbot Arena-style frontend for direct LLM comparison and executed load/stress testing for production deployment

- **Machine-Readable Government (MRG) · DGE Abu Dhabi**

- Designed and built a production LLM compliance-automation platform for government product-registration review, converting policy PDFs into machine-readable policy graphs and running automated compliance assessments over reviewable case workflows
- Architected a 4-stage multi-agent workflow (retrieval → taxonomy → compliance → decision) on the OpenAI Agents SDK, orchestrating end-to-end case adjudication with sandboxed, multi-turn tool-using agents and rule-guided decisions
- Engineered a hybrid retrieval pipeline pairing OpenAI Vector Store semantic search (high recall) with local policy-graph validation (high precision), enabling deterministic, cross-referenced policy applicability expansion per case
- Built a PDF → Markdown → policy-graph extraction pipeline with 3-stage LLM validation, preserving source-clause locators and resolving cross-policy replaces / cancels / amends relationships for auditable compliance checks
- Developed a comprehensive evaluation framework — retrieval, compliance, decision, graph-accuracy, and variance evals with LLM-based graders and exact-match validators — run through a containerized, sharded, resumable harness against golden datasets
- Led LLM observability with Langfuse (content-rich traces, prompt versioning) alongside an OpenTelemetry structural-telemetry stream to Azure App Insights / SigNoz, with automatic secret scrubbing and environment-gated content capture
- Deployed on Azure Kubernetes Service with Terraform IaC and Azure Service Bus-throttled worker dispatch, containerizing FastAPI / React services and achieving multi-tenant isolation via Postgres row-level security and workload identity
- Built the CI/CD pipeline (GitHub Actions + Azure Pipelines) enforcing Ruff / pytest, ESLint / Vitest, and Terraform / Checkov gates behind branch protection, and secured the platform with Microsoft Entra ID (OAuth2 + PKCE) browser auth, app-only bearer tokens for service clients, and CSRF-protected signed sessions

Boston Consulting Group, Data Scientist Intern

- Increased call transcription accuracy by analysing audio to identify language and tailor transcription strategies
- Developed a complete audio processing pipeline using VAD for noise reduction, Mel-Spectrogram features, and a TDNN-based language identification model
- Fine-tuned the Whisper model with hyper-parameter tuning for Indian languages, significantly improving transcription fidelity
- Engineered a QA module to flag recurring lines and timestamp discrepancies; leveraged GPT-3.5 and XLM-RoBERTa for business category classification

Mumbai, India
Oct 2023 – Apr 2024
7 months

Education

PG Diploma Indian Institute of Technology Kharagpur, Business Analytics

- Relevant Coursework: Statistical Learning, Deep Learning, Bayesian Inference, Categorical Data Analysis
- Editor-in-Chief of the PGDBA Annual AI and Analytics Magazine (AINA 4.0), leading a team of 6

Kharagpur, India
Aug 2022 – May 2024

MSc Chennai Mathematical Institute, Pure Mathematics

- Relevant Coursework: Measure Theory, Stochastic Process, Operator Theory, Ergodic Theory
- Thesis: Angle & Distance Estimates and Kazhdan Constants
- Full academic scholarship and tuition fee waiver

Chennai, India
Aug 2020 – June 2022

BSc Chennai Mathematical Institute, Mathematics and Theoretical Computer Science

- Relevant Coursework: Probability, Algebra, Topology, Manifolds, Riemannian Geometry, Algorithms
- Full academic scholarship and tuition fee waiver

Chennai, India
Aug 2017 – June 2020

Publications

PETRA: Transforming Web Text for Petroleum-Engineering Domain Adaptation

June 2026

K. Dubovikov, O. El Mansouri, H. Madmoun, Y. Li, S. Kumar, A. El Mir, S. Ghosh, **W. Bhattacharya**, A. Garcia-Garcia, O. Pandit, S. K. Sahu, F. Castanedo, L. Murray, M. Takac, S. Lahlou
arxiv.org/abs/2606.24346 (arXiv preprint (arXiv:2606.24346))

Assessing The Capabilities Of Large Language Models For Oil And Gas Industry Applications

Nov 2024

W. Bhattacharya, et al.

onepetro.org/SPEADIP/proceedings/24ADIP/24ADIP/D021S033R003/579034 (SPE Journal (SPE-229701-MS) — presented at ADIPEC 2024)

Awards

Olympiads – INMO, INPhO, INAO: Selected among India's top 30 students in the Mathematics (INMO), Physics (INPhO), and Astronomy (INAO) Olympiads; advanced to the national training camp at HBCSE, TIFR.

INSPIRE Scholarship: Government of India – DST. Awarded for outstanding academic performance (₹1.8L p.a. for 3 years).

JEE Mains – AIR 1018: Top 0.085 percentile out of 1.2 million candidates (2017).

JEE Advanced – AIR 1039: Top 0.6 percentile in JEE Advanced (2017).

CMI Full Scholarship: Full academic scholarship and tuition fee waiver for both the B.Sc. and the M.Sc., awarded by Chennai Mathematical Institute.

Leadership

Editor-in-Chief – AINA 4.0: PGDBA Annual AI and Analytics Magazine. Led a team of 6; featured an interview with Dr. Josh Starmer (StatQuest).

Olympiad Mentor – Cheenta Ganit Kendra: Mentored Mathematics and Physics Olympiad aspirants for four years (2017–2021).

Skills

Machine Learning & AI: LLMs, RAG, Fine-tuning (LoRA/PEFT), TTS, STT, NLP, Embeddings, Transformers, Reinforcement Learning

MLOps & Engineering: Docker, Kubernetes, FastAPI, Azure, Databricks, vLLM, Gradio, Load Testing, Production Deployment

Statistics & Mathematics: Time Series, Stochastic Processes, Bayesian Inference, Measure Theory, Statistical Learning

Programming: Python, R, Haskell, C++, JavaScript

Frameworks & Tools: PyTorch, LangChain, Hugging Face, verl, vLLM

Languages

English: Fluent

Bengali: Native

Hindi: Fluent

Spanish: Beginner

Interests

Research: World Models, Reinforcement Learning, LLMs, Speech AI, Quantitative Finance, Game Development

Hobbies: Football, Basketball, Badminton, Padel, Karting, Guitar